

An Exact Computer-Assisted Optimality Certificate for Ten Disks of Radii $1, 2, \dots, 10$ in a Circle

Computer-assisted proof manuscript

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Abstract

We study the smallest circular container for ten pairwise nonoverlapping closed disks with radii $1, 2, \dots, 10$. We prove, subject to the stated exact-arithmetic verifier trust boundary, that the optimum is the uniquely specified angle-closure root $22.000193012737 < R_* < 22.000193012738$. An exact upper witness has five mutually tangent wall disks of radii $6, \dots, 10$, and rational centers for the remaining five disks. The global lower bound does *not* follow from the five wall disks alone: they can fit into a circle of radius 21.9 in a different order. Instead, we include disk 5 in a six-disk core and cover every possible cyclic order with a 75-node rational radial tree. Thirty-four leaves are excluded by certified negative cycles and four leaves by a monotonicity barrier for the five-disk contact loop. The replay uses only integer and rational arithmetic, apart from untrusted discovery procedures.

1 Problem and exact result

For $1 \leq i \leq 10$, let disk i have radius $r_i = i$, center $p_i \in \mathbb{R}^2$, and radial center distance $s_i = \|p_i\|$. A feasible configuration in a container of radius R satisfies

$$s_i + i \leq R \quad (1 \leq i \leq 10), \quad \|p_i - p_j\| \geq i + j \quad (1 \leq i < j \leq 10). \quad (1)$$

Define $U = 22.00019301274$. Set

$$\beta_{ij}(R) = 2 \arcsin \sqrt{\frac{ij}{(R-i)(R-j)}} \quad (2)$$

for R in the domain on which the right side is real. Put

$$\mathcal{B}(R) = \beta_{10,8}(R) + \beta_{8,6}(R) + \beta_{6,7}(R) + \beta_{7,9}(R) + \beta_{9,10}(R). \quad (3)$$

Theorem 1 (Exact global optimality). *There is a unique number*

$$R_0 \in [22.000193012737, 22.000193012738]$$

for which $\mathcal{B}(R_0) = 2\pi$. The minimum container radius for radii $1, \dots, 10$ is exactly R_0 . In particular,

$$\boxed{R_* = R_0 = 22.00019301273737054809 \dots}.$$

The decimal expansion is for orientation; Equation (3) and the rational isolating interval specify the exact real number.

2 Angle inequalities and exact finite certificates

If two center distances are $x = s_i > 0$ and $y = s_j > 0$, their pairwise non-overlap imposes the smaller central-angle bound

$$\phi_{ij}(x, y) = \arccos\left(\frac{x^2 + y^2 - (i + j)^2}{2xy}\right), \quad (4)$$

whenever $|x - y| < i + j \leq x + y$; endpoint cases follow by continuity. The directed gap in any chosen cyclic order is at least this necessary smaller angle.

Lemma 2 (Four-corner cosine bound). *Fix $d > 0$ and a positive rectangle $[a, b] \times [c, e]$. The maximum of*

$$f(x, y) = \frac{x^2 + y^2 - d^2}{2xy}$$

is attained at one of its four corners.

Proof. For fixed $y > 0$, the derivative $\partial f / \partial x = (x^2 - y^2 + d^2) / (2yx^2)$ changes sign, if at all, only from negative to positive. Thus its interior stationary point is never a strict maximum. Maximize first over x , then over y . $\square$

On a rational radial box $x \in [\ell_i, u_i]$, $y \in [\ell_j, u_j]$, we compute the exact four-corner maximum c_{ij}^+ . A rational angle tick $q_{ij} = k_{ij}/10^6$ is accepted only when a rational Taylor *lower* bound for $\cos q_{ij}$ exceeds c_{ij}^+ ; then $\phi_{ij} \geq q_{ij}$. All positive-center assumptions and each acceptance test are checked again by the verifier.

For a fixed linear lift of a cyclic order $\theta_0 \leq \dots \leq \theta_{m-1} < \theta_0 + 2\pi$, every pair of positions $a < b$ satisfies

$$q_{i_a i_b} \leq \theta_b - \theta_a \leq 2\pi_+ - q_{i_a i_b}, \quad \pi < \pi_+ = 3.141592653590. \quad (5)$$

These are rational difference constraints $\theta_v - \theta_u \leq w_{uv}$.

Lemma 3 (Negative-cycle exclusion). *If the difference-constraint digraph contains a directed cycle C for which $\sum_{e \in C} w_e < 0$, its corresponding cyclic order cannot occur in a feasible disk packing.*

Proof. Sum the constraints along C . The angle differences telescope to zero, forcing the contradiction $0 \leq \sum_{e \in C} w_e < 0$. $\square$

3 An exact ten-disk upper witness

Lemma 4 (Exact angle root). *The equation $\mathcal{B}(R) = 2\pi$ has exactly one root in $J_R = [22.000193012737, 22.000193012738]$, and $R_0 < U$.*

Proof. Each summand β_{ij} is strictly decreasing in R on its domain:

$$\beta'_{ij}(R) = -\left(\frac{1}{R-i} + \frac{1}{R-j}\right) \tan \frac{\beta_{ij}(R)}{2} < 0.$$

The exact verifier establishes $\mathcal{B}(\inf J_R) > 2\pi$ and $\mathcal{B}(\sup J_R) < 2\pi$ by rational enclosures for square roots, π , and arctangents. Continuity and strict monotonicity give the unique root. The upper endpoint of J_R is strictly below U . $\square$

Set the wall core order to

$$(10, 8, 6, 7, 9). \quad (6)$$

Put $p_{10} = (R_0 - 10, 0)$. For the subsequent disks, define their polar angles recursively by adding the successive tangency angles from (2), and set each radial distance to $R_0 - i$. The closing equation makes the final pair (9, 10) tangent as well. The other centers are the following *exact rational points*:

$$\begin{aligned} p_1 &= (-8.901, 19.006), & p_2 &= (-15.551, -12.425), \\ p_3 &= (12.838, 13.952), & p_4 &= (11.103, -14.109), \\ p_5 &= (-3.357, 1.306). \end{aligned} \quad (7)$$

Lemma 5 (Witness feasibility). *The polar construction (6) together with the five rational centers (7) satisfies all constraints (1) at $R = R_0$.*

Proof. Five consecutive pairs in (6) are tangent by the law of cosines and the defining angular equation. The other 40 unordered pairs have strictly positive squared-distance slack according to exact rational interval evaluations throughout $R \in J_R$. The five loose disks satisfy strict containment; the core disks are on the wall. The smallest certified lower bound for a squared-distance slack is larger than 3.48 (for disks 5 and 7). $\square$

The construction proves $R_* \leq R_0$. Figure 1 shows the arrangement using *rounded* coordinates only.

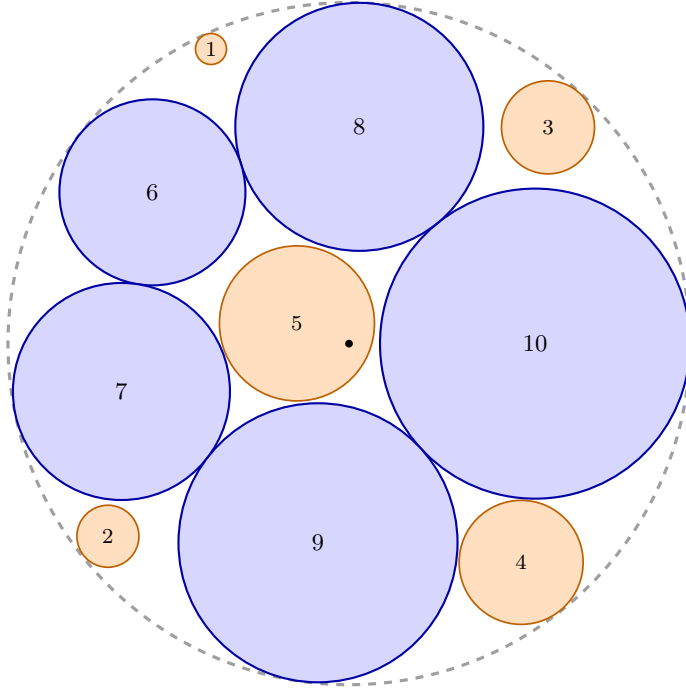


Figure 1: Illustrative optimal ten-disk configuration. Blue: five mutually contacting wall disks. Orange: five rationally placed additional disks. Neither feasibility nor tangency is inferred from the rounded drawing.

4 A universal radial cover for the six-disk lower bound

The lower bound is proved using only disks 5, 6, 7, 8, 9, 10; disks 1–4 can be dropped from an impossible-packing argument. Suppose a ten-disk packing existed with $R < R_0$. Then $R < U$,

and for each $i \in \{5, \dots, 10\}$, containment and separation imply

$$\max \left(0, \max_{j \neq i} (i + 2j - U) \right) \leq s_i \leq U - i, \quad (j = 5, \dots, 10). \quad (8)$$

These give strictly positive lower radii for all six selected centers. We bisect these rational intervals at rational midpoints and propagate $s_i + s_j \geq i + j$. Every pair of children covers the parent, so the tree covers *all* radial vectors of hypothetical packings, not a sample.

The six center directions admit $(6 - 1)!/2 = 60$ possible cyclic orders up to rotation and reflection, rooted at disk 10. On each terminal radial box, rational angle ticks are checked for all 15 pairs, and Lemma 3 closes every incompatible order unless the main-order geometric terminal rule below applies.

Lemma 6 (Wall-angle monotonicity on a certified box). *Let the projection of the six-disk center order to the five labels $\{6, 7, 8, 9, 10\}$ be exactly $(10, 8, 6, 7, 9)$. Suppose the radial box, with every selected s_i allowed to increase up to $U - i$, satisfies for every consecutive pair (i, j) in this five-cycle*

$$\begin{aligned} \ell_i + \ell_j &> i + j, \\ \max\{|\ell_i - (U - j)|, |(U - i) - \ell_j|\} &< i + j, \\ \ell_i^2 - (U - j)^2 + (i + j)^2 &> 0, \\ \ell_j^2 - (U - i)^2 + (i + j)^2 &> 0. \end{aligned}$$

Then no packing in the box is possible for any $R < R_0$.

Proof. On the indicated rectangles, the necessary angle (4) is in its smooth triangle-inequality domain, and

$$\frac{\partial \phi_{ij}}{\partial s_i} = -\frac{s_i^2 - s_j^2 + (i + j)^2}{2s_i^2 s_j \sin \phi_{ij}} < 0,$$

with the analogous derivative in s_j . Thus replacing the five radial coordinates, *only in the angle formula*, by their containment maxima $R - i$ can only decrease the sum of the five required angles. Therefore any feasible packing would satisfy

$$2\pi \geq \sum_{(i,j) \text{ in five-cycle}} \phi_{ij}(s_i, s_j) \geq \mathcal{B}(R) > \mathcal{B}(R_0) = 2\pi,$$

where strictness uses $R < R_0$ and the decreasing property of $\mathcal{B}$. This is impossible. No physical disk movement is asserted. $\square$

Proposition 7 (Exhaustive rational terminal certificate). *There exists a binary tree of rational radial boxes with the exact statistics in Table 1. Every terminal box either excludes all sixty orders using certified negative cycles, or satisfies the hypotheses of Lemma 6 for every order not already excluded. Consequently no disks of radii 5, 6, 7, 8, 9, 10 can fit at $R < R_0$.*

Proof. The accompanying JSON certificate gives the splitting disk and exact rational cut at each internal node. Every terminal node stores rational angle ticks and one simple negative cycle per excluded linearized order. The verifier checks every cosine Taylor witness, edge orientation, cycle closure, and strictly negative exact rational edge sum. At the other four terminal nodes it independently checks every inequality of Lemma 6, and verifies that any order lacking a negative cycle has five-disk projection $(10, 8, 6, 7, 9)$. The verifier recursively reconstructs every box and checks the complete set of sixty orders, without trusting the discovery program or the proposed classifications. $\square$

Table 1: Exact replay of the six-disk lower-bound certificate.

Property	Verified count
Total tree nodes	75
Rational midpoint splits	37
Terminal boxes with all orders excluded by negative cycles	34
Terminal boxes requiring the wall-angle monotonicity rule	4
Unresolved terminal boxes	0
Maximum depth	30

5 Why five largest disks alone do not suffice

The five disks of radii 6, 7, 8, 9, 10 can in fact fit into a container of radius $219/10 = 21.9 < R_0$ in a *different* cyclic order. An exact rational witness, in the order (10, 6, 9, 8, 7), is

$$\begin{aligned}
p_{10} &= (11.88, 0), & p_6 &= (5.26, 14.983), \\
p_9 &= (-9.029, 9.185), & p_8 &= (-11.333, -8.013), \\
p_7 &= (2.593, -14.652).
\end{aligned}$$

All containment and ten pairwise separation inequalities for these five disks hold strictly, as can be checked directly by squaring the rational coordinates. This demonstrates why incorporating disk 5 into the *lower-bound* case analysis is indispensable to this proof strategy.

6 Proof of the main theorem and reproducibility

Proof of Theorem 1. Lemma 4 supplies the exact critical root. Lemma 5 proves $R_* \leq R_0$. Proposition 7 excludes even the six-disk subset of radii 5, $\dots$, 10 at every $R < R_0$. Hence the full ten-disk packing cannot exist there, proving $R_* \geq R_0$. $\square$

The certificate and independent replay components are supplied alongside this manuscript. From their common directory run

```
python3 certify_integer_ten_global.py
```

This invokes `verify_ten_upper.py` for the angle root and exact witness and `verify_ten_global_tree.py` for the global finite cover. The verifier uses Python integers, `Fraction`, integer square roots, and rational Taylor enclosures for trigonometric quantities; candidate angle ticks may be found using floating point *before* verification. Optimized Python modes `-O` and `-OO` are rejected because the checkers use `assert`. The mathematical soundness claim remains conditional on the correctness of these programs, the Python runtime and standard-library semantics, and the certificate files. It is not an end-to-end proof-assistant formalization, and has not received independent peer review.

References

- [1] E. Specht, *The best known packings of unequal circles with integer radii in a circle*, Packomania, updated September 10, 2026, <https://packomania.com/ccin/ccin.html>.